# Exactly Normalized Pose Distributions by Construction: Hierarchical Partition Trees over SO(3) and SE(3)

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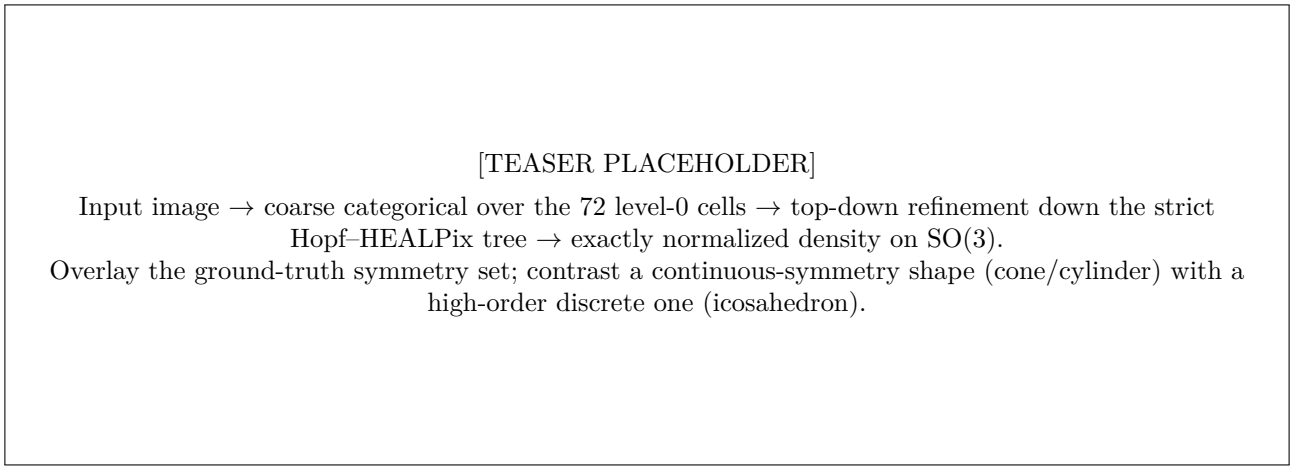



Figure 1. NeST estimates a pose distribution without a retrieval grid or a sampled partition function. The distribution is a tree-structured softmax over a strict hierarchical equivolumetric partition of a pose group; the log-likelihood of any pose is read off by scoring only the $L{+}1$ nodes on its path. We instantiate the framework on SO(3) (shown) and on SE(3). [TODO: replace with final figure]

## Abstract

Estimating the full distribution of an object's pose—rather than a single estimate—is essential when symmetry, self-occlusion, or sensor noise make the pose ambiguous. The dominant non-parametric approach, Implicit-PDF, represents the distribution implicitly as an energy over SO(3) and must evaluate a dense equivolumetric grid of millions of rotations—both to approximate the normalizer during training and to recover a normalized density at inference. A hierarchical variant, SpyroPose, keeps the energy formulation but estimates the partition function with importance sampling during training and evaluates a sparse coarse-to-fine tree at inference, and extends to SE(3) with an energy pyramid. In both cases the distribution is an energy whose cost is tied to how many poses are scored.

We revisit the problem from the opposite direction, and in general. For any pose group with a strict hierarchical equivolumetric partition—one in which every cell splits into a fixed number of disjoint equal-volume children—a distribution over the leaves is a tree-structured softmax: a categorical over the coarsest cells and, at every expanded node, a categorical over its children. This is an exactly normalized probability by design, independent of the group: no retrieval grid and no sampled partition function are ever needed, the log-likelihood of any pose is read off in closed form from the $L{+}1$ nodes on its path (cost $O(L)$ rather than $O(N)$ for $N$ cells), and a uniform predictor provably scores $-\ln V_G$ at every resolution—a data-free self-test of the whole likelihood. We instantiate the framework on SO(3) via the Hopf–HEALPix grid (eight children per node) and on SE(3) via the Cartesian product of that rotation tree with a bounded equivolumetric translation octree (sixty-four children per node).

A lightweight cross-attention head on an ImageNet-pretrained vision transformer realizes the construction, zero-initialized so training starts exactly at the uniform distribution, trained by maximum likelihood of the single ground-truth pose.

On SYMSOL I a single joint model reaches 7.71 average log-likelihood, surpassing the strongest RGB-only baseline while reading any rotation's exact likelihood from a seven-node path instead of a million-cell grid, with near-grid-floor beam-MAP accuracy except on the high-order icosahedral object, whose sixty equivalent modes are the hardest to resolve. The SE(3) instance is exactly normalized by the same construction—its uniform predictor provably scores $-\ln(\pi^2 V_{\text{box}})$, which we verify numerically to machine precision—and we set up its evaluation, together with SYMSOL II and the real-image BOP datasets T-LESS and YCB-Video, under one protocol spanning both pose groups. Code and the preprocessing pipeline will be released.

## 1. Introduction

Recovering the 6-DoF pose of an object—its orientation in SO(3) and its position in $\mathbb{R}^3$, jointly an element of SE(3)—from a single image is a core problem in computer vision and robotics. In many settings the pose is not uniquely determined by the image. A textureless cube looks identical under its 24 rotational symmetries and a cylinder under a continuous one-parameter family; depth is ambiguous under perspective; and self-occlusion or a missing distinguishing feature can make otherwise distinct poses indistinguishable. A method that returns a single pose discards this structure; a method that returns a distribution over the pose group preserves it, and lets downstream consumers—grasp planners, trackers, sensor-fusion pipelines—reason about the ambiguity explicitly [6, 17]. The difficulty is normalization: a usable distribution must integrate to one over the group, and for the expressive non-parametric models this is exactly the expensive part, on SO(3) and more so on SE(3).

The most expressive way to represent such a distribution is non-parametrically. Implicit-PDF (IPDF) [17] does so with an implicit function: a network maps an image embedding and a candidate rotation to an unnormalized score, and the distribution is recovered by normalizing these scores over an equivolumetric grid of SO(3). This is maximally flexible and trains from a single ground-truth rotation per image, but it inherits two costs from the grid. First, obtaining a normalized density requires evaluating the network at every grid point—roughly 2.4M rotations at the resolution used for evaluation—so a faithful distribution is expensive and its fidelity is capped by the grid spacing. Second, during training the normalizer is only approximated, by querying a random or uniform coverage of SO(3) together with the ground truth. SpyroPose [6] makes the grid hierarchical and replaces the uniform coverage with importance sampling, reducing the number of evaluations by orders of magnitude and extending the idea to SE(3); it trains an energy per resolution with a contrastive objective whose partition function is an importance-sampling estimate, and at inference expands the top-$k$ cells level by level and renormalizes their children. Its inference distribution is properly normalized, but it is obtained by a multi-query sparse search and its fine-level density depends on the expansion width $k$.

Our approach. We ask whether the normalizer can be removed from the problem entirely rather than approximated more cheaply. Our observation is simple: the same Hopf–HEALPix construction that IPDF and SpyroPose use for its equivolumetric property is also strictly hierarchical—doubling the grid resolution splits every cell into exactly eight disjoint children, forming a tree. On a tree, a normalized distribution over the leaves can be written as a product of per-node categoricals (a tree-structured softmax [16]): a categorical over the 72 cells at the coarsest level, and at every node a categorical over its 8 children. Because each factor is a softmax, the product is an exactly normalized probability by construction—at every resolution, with no grid evaluation and no sampled partition function. The density at a rotation $R$ is

$$p(R) \;=\; P\big(\text{finest cell of } R\big) \cdot \frac{N}{\pi^2}, \qquad (1)$$

and $\log P(\text{cell})$ is the sum of the log-categoricals along the single root-to-leaf path that contains $R$. Evaluating the likelihood of any rotation therefore costs $O(L)$ node scorings for a grid of $N = 72 \cdot 8^L$ cells, rather than the $O(N)$ of a retrieval grid.

This argument never used anything special about rotations: it holds for any pose group with a strict hierarchical equivolumetric partition. We therefore state the method as a general framework (Sec. 3.1) and instantiate it twice: on SO(3) with the Hopf–HEALPix tree above, and on SE(3) by taking the Cartesian product of that rotation tree with a bounded equivolumetric translation octree (Sec. 3.7), under which a joint cell factorizes into a rotation path and a translation path and the exact normalization and $O(L)$ likelihood carry over unchanged.

We instantiate the tree with a compact head on top of a vision transformer [3]. A set of learned queries

cross-attends to the patch tokens to produce the coarsest categorical; a single shared child scorer, conditioned on each child's geometry relative to its parent, produces the 8-way categorical at every node. The output layer is zero-initialized, so the model starts from the uniform distribution and the training log-likelihood starts at the exact uniform floor $-\ln \pi^2$. Training is maximum likelihood of the single rendered ground-truth rotation—no symmetry annotations, matching the IPDF protocol.

Contributions.

- A general framework for pose-distribution estimation that is exactly normalized by construction: a tree-structured softmax over any strict hierarchical equivolumetric partition of a pose group. It needs neither a retrieval grid nor a sampled partition function, the exact log-likelihood of any pose is read off in $O(L)$ from a single path, and a uniform predictor provably scores $-\ln V_G$—a data-free self-test (Sec. 3.1).
- Two instantiations: SO(3) via the Hopf–HEALPix tree (8 children per node), and SE(3) via the Cartesian product of that tree with a bounded equivolumetric translation octree (64 joint children per node), which keeps the exact normalization and $O(L)$ likelihood and adds only a detector-sized translation box (Sec. 3.7).
- A lightweight cross-attention head on a frozen-init vision transformer that realizes the construction, trained by single-ground-truth maximum likelihood, with an $O(bL)$ beam-search MAP decoder.
- An evaluation on SYMSOL I under both the distribution and pose metrics, with analysis and ablations (Sec. 4): a single joint model reaches 7.71 average log-likelihood, above the strongest RGB-only baseline (7.33, trained per shape), while reading the exact likelihood of any rotation from a single 7-node path rather than scoring a million-cell grid; its one weak case is the high-order icosahedral object, whose 60 equivalent modes must be covered at once. The SE(3) instance is exactly normalized by the same construction, which we verify numerically; we define a single protocol that carries it, together with SYMSOL II and the real-image BOP datasets T-LESS and YCB-Video, across both pose groups (Sec. 4.7).

## 2. Related Work

Parametric distributions on SO(3). A long line of work models rotational uncertainty with a parametric density whose parameters are regressed from the image. The von Mises [19], Bingham [2, 4], and matrix-Fisher [15] distributions are the most common choices; multimodality is obtained by mixing components. These models are compact and have closed-form statistics, but their normalizing constants require lookup tables or series approximations, and a fixed number of modes limits how faithfully they can represent high-order or continuous symmetries [17]. The Rotation Laplace distribution [25] improves robustness within the parametric family.

Non-parametric and implicit distributions. IPDF [17] represents the distribution implicitly as a network that scores (image, rotation) pairs, normalized over an equivolumetric Hopf–HEALPix grid [5, 24], and introduced the SYMSOL benchmark we use. It is maximally expressive and trains from a single pose, but a normalized density requires a dense grid evaluation at inference. HyperPosePDF [8] predicts the weights of such an implicit function with a hypernetwork. Image2Sphere [10] maps image features onto the sphere and applies SO(3)-equivariant layers before scoring, improving generalization at a lower resolution. AQuaMaM [1] is autoregressive over a quaternion manifold. Our method is non-parametric in the same sense—arbitrarily many modes at arbitrary resolution—but differs in that the distribution is normalized analytically rather than by grid evaluation or sampling.

Hierarchical grids and the partition function. SpyroPose [6] is the closest prior work: it uses the same hierarchical Hopf–HEALPix grid, extended to an SE(3) pyramid, and is the state of the art on SYMSOL rotation distributions. It learns an energy (unnormalized log-likelihood) at each resolution with an InfoNCE/NCE objective [21], using the hierarchy to draw hard negatives for an importance-sampling estimate of the partition function during training. Notably, SpyroPose already derives a product-of-sibling-softmaxes form (its Eq. 6), a "generative coarse-to-fine Markov chain," but uses it only as the importance-sampling proposal during training; at inference it expands the top-$k$ cells level by level and renormalizes their children, so the fine density, while normalized, depends on the expansion width $k$. Our contribution is to make that factored form the model itself: we train the tree-structured softmax directly by exact maximum likelihood, with no negatives and no partition-function estimate, so the exact log-likelihood of a rotation comes from a single root-to-leaf path and a full-depth density is independent of any $k$. We further replace the per-level MLPs with a single, shared, geometry-conditioned child scorer. The construction connects to hierarchical softmax [16] and to autoregressive models over dis-

cretized manifolds [1], with the tree depth playing the role of the autoregressive axis.

Generative and equivariant approaches. Recent work models pose distributions with normalizing flows on SO(3) [12], score-based diffusion on SE(3) [9], and direct prediction of Wigner-D harmonic coefficients [11]. Flows and diffusion produce samples and (for flows) densities but require integration or iterative sampling to evaluate a likelihood; our density is available in closed form in a single forward pass per path.

Benchmarks. Our protocol spans both pose groups. For rotation we use SYMSOL [17] (distributions on synthetic symmetric solids). For full SE(3) we target the real-image objects T-LESS [7] and YCB-Video [23] under the BOP [18] protocol—the standard testbeds for pose estimation under symmetry and clutter—following the detector-conditioned, translation-box setup under which SpyroPose reports SE(3) log-likelihoods [6]. YCB-Video additionally admits point-based pose metrics (ADD(-S)) [23], letting a distribution model be scored against single-pose regressors.

# 3. Method

We cast pose-distribution estimation as a tree-structured softmax on a strict hierarchical equivolumetric partition of the pose group. Sec. 3.1 states the construction for a general group: any such partition induces an exactly-normalized density whose log-likelihood is read off in $O(L)$ from a single root-to-leaf path, with no normalizing integral to approximate. We then instantiate it on SO(3) (Sec. 3.2, the Hopf–HEALPix tree) and on SE(3) (Sec. 3.7, the Cartesian product of the rotation tree with a bounded translation octree), realize both with a single cross-attention head on a vision transformer (Sec. 3.4), train by single-ground-truth maximum likelihood (Sec. 3.5), and decode a MAP pose by beam search (Sec. 3.6).

## 3.1. General framework

Let $G$ be a group (or homogeneous space) carrying an invariant measure of finite total volume $V_G$. Suppose $G$ admits a strict hierarchical equivolumetric partition: a set of $N_0$ root cells of equal volume, in which every cell is subdivided into exactly $b$ disjoint children of equal volume, so level $L$ has $N(L) = N_0\, b^L$ equal-volume cells and each finest cell is identified by a unique path $(c_0, c_1, \dots, c_L)$ with $c_0 \in \{0, \dots, N_0{-}1\}$ and $c_l \in \{0, \dots, b{-}1\}$. On such a tree a distribution over the finest cells is a product of per-node categoricals: a root categorical $\pi_0 \in \Delta^{N_0-1}$ and, at every internal node $n$, a categorical $\pi_n \in \Delta^{b-1}$ over its children,

$$P(c_0, \dots, c_L) \;=\; \pi_0[c_0] \prod_{l=1}^{L} \pi_{n_{l-1}}[c_l]. \qquad (2)$$

Every factor is a normalized categorical, so $\sum P = 1$ exactly at every depth, with no partition function to sample or integrate. The induced density is piecewise-constant with value $p(x) = P(\mathrm{cell}_L(x))\, N(L)/V_G$, so a uniform predictor gives $p \equiv 1/V_G$ and average log-likelihood $-\ln V_G$ at every level—an exact, data-free self-test of the entire likelihood path. Evaluating $\log p(x)$ for any $x$ touches only the $L{+}1$ nodes on its path ($O(L)$), versus the $O(N)$ evaluation a retrieval grid needs for a normalized density, and a MAP estimate follows from an $O(bL)$ beam search. The remaining design choices are group-specific: the partition (hence $N_0$ and $b$) and how a network conditions each node's categorical. We give two instances: SO(3) ($N_0{=}72$, $b{=}8$, $V_G{=}\pi^2$) and SE(3) ($b{=}64$, $V_G{=}\pi^2{\cdot}V_{\mathrm{box}}$).

## 3.2. Instantiation I: SO(3)

We use the equivolumetric grid of Yershova et al. [24], as adopted by IPDF [17]: the 2-sphere base point is tiled by HEALPix [5] in the nested scheme ($12\,\mathrm{nside}^2$ equal-area pixels) and the Hopf fibre by $6\,\mathrm{nside}$ equal angles, giving

$$N(l) \;=\; 72 \cdot 8^{\,l} \qquad (3)$$

equal-volume cells at level $l$, with total volume $V_{\mathrm{SO(3)}} = \pi^2$. The key property we exploit, beyond equal volume, is that the grid is a strict tree: doubling nside splits every HEALPix pixel into 4 (nested) and every fibre bin into 2, so each cell has exactly 8 disjoint children. A finest cell at level $L$ is therefore identified by a unique path $(c_0, c_1, \dots, c_L)$, where $c_0 \in \{0, \dots, 71\}$ indexes the coarsest cell and each $c_l \in \{0, \dots, 7\}$ selects a child. We map between a rotation and its cell via the Hopf lift and its inverse, so the path of any rotation is obtained directly without enumerating the grid.

## 3.3. Tree-structured softmax distribution

We define a distribution over the $N(L)$ finest cells as a product of per-node categoricals. Let $\pi_0 \in \Delta^{71}$ be a categorical over the 72 coarsest cells, and for every internal node $n$ let $\pi_n \in \Delta^7$ be a categorical over its 8 children. The probability of a finest cell with path $(c_0, \dots, c_L)$ is

$$P(c_0, \dots, c_L) \;=\; \pi_0[c_0] \prod_{l=1}^{L} \pi_{n_{l-1}}[c_l], \qquad (4)$$

where $n_{l-1}$ is the node reached after $l-1$ steps. Because every factor is a normalized categorical, $\sum P = 1$

exactly, at every depth $L$, with no normalizing integral to approximate. The continuous density assigns each cell its probability spread over its volume,

$$p(R) \;=\; P\big(\mathrm{cell}_L(R)\big)\cdot\frac{N(L)}{\pi^2}, \qquad \log p(R) = \log P + \ln\frac{N(L)}{\pi^2} \tag{5}$$

A uniform predictor ($\pi_0$ and all $\pi_n$ uniform) gives $p(R) = 1/\pi^2$ and average log-likelihood $-\ln\pi^2 \approx -2.29$ nats at every level—an exact self-test of the whole likelihood path, independent of data or training. Crucially, computing $\log p(R)$ for any $R$ touches only the $L{+}1$ nodes on its path: cost $O(L)$, versus the $O(N)$ grid evaluation a retrieval model needs for a normalized density.

### 3.4. Network

Backbone. We use an ImageNet-pretrained ViT-B/16 (augreg2_in21k_ft_in1k) [3, 22] at $224 \times 224$, taking its patch tokens $\mathbf{Z} \in \mathbb{R}^{P\times D}$ ($D = 768$) after dropping the prefix tokens, and fine-tune it end to end. Note this departs from the ResNet-50 used by IPDF [17] and SpyroPose [6]; because the backbone is a confound when comparing distribution formulations, we include a backbone-matched comparison in Sec. 4.6 (Tab. 13).

Coarsest categorical. A set of 72 learned query embeddings cross-attends to the patch tokens through a small transformer (two layers, eight heads, query-only updates with an MLP) and is projected to one logit each; a log-softmax over the 72 logits gives $\log\pi_0$.

Child scorer. A single scorer, shared across all nodes and levels, produces the 8-way categorical at every expanded node. Its 8 queries are built from the geometry of the node's children: for each child we take its offset from the parent cell centre as a rotation vector, normalized by the characteristic cell size $(\,\pi^2/N(l)\,)^{1/3}$ at that level; we Fourier-encode this 3-D offset and the flattened 9-D parent rotation (positional encoding with $n_{\text{freq}} = 3$, following [14]); we concatenate the two and add a learned level embedding. These 8 queries cross-attend to the same patch tokens through an identical two-layer, eight-head block and are projected and log-softmaxed to $\log\pi_n$. Sharing the scorer across levels keeps the parameter count independent of depth $L$ and lets the same conditioning generalize across resolutions. The output linear layers of both scorers are zero-initialized, so every categorical starts uniform and training begins exactly at the $-\ln\pi^2$ floor.

### 3.5. Training

We train by maximum likelihood of the single rendered ground-truth rotation, exactly as in IPDF—no symmetry annotations and no equivalent set are used at training time, since that would give our model information the baselines lack. For a training pair $(I, R)$ we compute the path of $R$ at level $L$ and minimize the negative log-likelihood

$$\mathcal{L} \;=\; -\Big(\log\pi_0[c_0] + \sum_{l=1}^{L}\log\pi_{n_{l-1}}[c_l]\Big), \tag{6}$$

i.e. $-\log P$ from Eq. (4); the constant $\ln(N/\pi^2)$ is added back only for reporting. Only the $L{+}1$ nodes on the ground-truth path are scored per image, so the backward pass is $O(L)$.

Optimization. We train end to end in fp32 (the finest cells are too small for bf16) with AdamW [13] (weight decay 0.05), learning rate $10^{-4}$ for the head and $10^{-5}$ for the backbone, a one-cycle schedule [20] with 5% warmup, batch size 64, gradient clipping at 1.0, for 60k steps. The data pipeline precomputes each sample's path and the per-child geometry the scorer consumes, so the network never touches the grid library at train time.

Exact in-plane roll augmentation. Single–ground-truth training has a failure mode that is specific to symmetric objects. The $K$ symmetric copies of a pose are indistinguishable from the image, so no model can place its mass on the particular copy that was rendered from the image alone; the single-rotation likelihood is therefore bounded, and a model that appears to beat that bound on the training set can only have memorized which copy each training view was paired with. Empirically this is exactly what happens: the held-out likelihood peaks and then degrades as training proceeds (Tab. 8), worst on the high-order symmetries. We remove it with an augmentation that is exact, not approximate. SYMSOL views are centred, rendered on a uniform background with the object inside the inscribed circle, so rotating the image in-plane by an angle $\theta$ about its centre is identical to rolling the camera about its optical axis—a rigid transform of the pose. The rolled frame is thus a genuine, correctly labelled render whose ground-truth rotation is $R_z(-\theta)\,R$ (the sign and axis are fixed empirically by scoring the four candidates $R_z(\pm\theta)R$, $R\,R_z(\pm\theta)$ under a trained model and keeping the one that preserves the likelihood). We draw $\theta \sim \mathcal{U}[0, 2\pi)$ per sample in the data loader, roll the frame—filling the exposed corners with the background colour—and recompute the path of $R_z(-\theta)\,R$ on the fly. This preserves the single-GT rule exactly (one rotation per image, merely relabelled) while making almost every view unique, so there is no

fixed copy assignment left to memorize. The relabel is a cell-size–normalized transform of the shared scorer's inputs, which is also why a model trained at one grid level transfers to a finer level with no retraining (Tab. 6).

### 3.6. Decoding a MAP rotation

For the point-estimate metric we recover an approximate MAP rotation by a top-down beam search over the tree. We keep the top-$b$ coarsest cells by $\log \pi_0$, expand each to its 8 children, score them with the shared child scorer, keep the top-$b$ of the resulting cells by cumulative log-probability, and repeat to level $L$; the highest-scoring leaf's cell-centre rotation is the estimate. We use beam width $b = 8$. Because scoring is exact and the tree is strict, this visits only $O(bL)$ nodes.

### 3.7. Instantiation II: SE(3)

SE(3) is unbounded in translation, so a strict equivolumetric partition requires a bounded translation box. We take the Cartesian product of the rotation tree of Sec. 3.2 with an equivolumetric octree over that box, exactly as the general framework of Sec. 3.1 prescribes; the tree-softmax, the $O(L)$ path likelihood and the beam decoder carry over unchanged.

Translation octree. Following SpyroPose [6], we parameterize the box by an affine map of a normalized unit cube $g \in [-\frac{1}{2}, \frac{1}{2}]^3$ into the camera frame, $t = \hat{t} + A\,g$, where $\hat{t}$ is a coarse position estimate (from a detector) and $A$ encodes the extent (in-plane $\sim d$ for an object of diameter $d$, enlarged along the view direction to absorb depth ambiguity). The octree subdivides the cube into the regular $2^l \times 2^l \times 2^l$ grid, $N_{\text{trans}}(l) = 8^l$ cells; because $g \mapsto \hat{t} + Ag$ is affine it preserves equal volume, so every translation cell has volume $|\det A|/8^l$ and the box has volume $V_{\text{box}} = |\det A|$.

Product tree. A joint cell factorizes into a rotation path and a translation path. The rotation root supplies the level-0 categorical over 72 cells (the translation root is a single cell); at each subsequent level a node has the full $b$=64 product of its 8 rotation children and 8 translation octants, with child index $c_{\text{rot}} \cdot 8 + c_{\text{oct}}$, until one half reaches its depth, after which the node is 8-way—so the rotation and translation depths $L_r, L_t$ are independent and can be tuned to the degrees-versus-millimetres precision each needs. Using the joint 64-way categorical rather than independent rotation and translation softmaxes lets the model couple apparent rotation and translation, which perspective makes dependent. The grid has $N = 72 \cdot 8^{L_r} \cdot 8^{L_t}$ cells and the density is

$$p(R,t) \;=\; P\big(\text{cell}(R,t)\big) \cdot \frac{N}{\pi^2 \cdot V_{\text{box}}}, \qquad (7)$$

so a uniform predictor scores $-\ln(\pi^2\, V_{\text{box}})$ at every level; we verify this identity numerically to machine precision, together with octree cell/path round-trips and the equality of the tree-softmax with the flat joint categorical, as a data-free self-test of the SE(3) construction (Appendix F). A pose whose translation leaves the box has zero density and infinite NLL, so the box must be sized conservatively from the detection and the in-box coverage reported; this is the one place the method depends on an external detector, as in [6]. The same cross-attention head (Sec. 3.4) realizes both instances—the child scorer simply emits up to 64-way logits—so training (Sec. 3.5) and decoding (Sec. 3.6) are identical across SO(3) and SE(3).

## 4. Experiments

### 4.1. Setup

Benchmark and protocol. We evaluate on SYMSOL [17]. SYMSOL I contains five textureless symmetric solids—tetrahedron, cube, icosahedron, cone, and cylinder—with discrete and continuous rotational symmetries; SYMSOL II contains three marked shapes (sphX, cylO, tetX) whose visible markers induce view-dependent (dynamic) ambiguity. We follow the IPDF protocol exactly: a single model is trained jointly over the five SYMSOL I shapes, supervised with only the single rendered rotation per image and no symmetry labels; at test time the full set of equivalent ground-truth rotations is used solely for evaluation.

Metrics. We report two families. (i) The distribution metric is the average log-likelihood of the ground-truth rotations, which for an image with equivalent set $\{R_k\}$ is $\frac{1}{K}\sum_k \log p(R_k \mid I)$; this is invariant to whether one or all equivalent annotations are available [17] and a uniform distribution scores $-\ln \pi^2 \approx -2.29$. (ii) The pose metric is the geodesic error of the beam-search MAP estimate to the nearest equivalent ground truth, summarized by median/mean error and accuracy at $2°/5°/10°$.

Discretization ceiling. Because our density is piecewise-constant on cells, the average log-likelihood is upper-bounded, for a given level $L$, by placing all mass uniformly on the occupied finest cells of the symmetry set: $\log\big(N(L)/(n_{\text{occ}} \cdot \pi^2)\big)$. We report this ceiling per shape and level in Tab. 1 so that model numbers can be read against what the grid resolution

Table 1. Discretization ceiling (max attainable average log-likelihood, nats) of the cell distribution per shape and level $L$, from the occupied finest cells of the symmetry set. Reference, not a result.

| $L$ | tet. | cube | ico. | cone | cyl. |
|---|---|---|---|---|---|
| 3 | 5.74 | 5.05 | 4.13 | 4.35 | 3.66 |
| 4 | 7.82 | 7.13 | 6.21 | 5.74 | 5.05 |
| 5 | 9.90 | 9.21 | 8.29 | 7.13 | 6.43 |
| 6 | 11.98 | 11.29 | 10.37 | 8.58 | 7.89 |

allows; continuous symmetries (cone, cylinder) have the lowest ceilings because their one-parameter orbit occupies many cells.

Implementation. The headline results (Tabs. 2, 4 and 5) use $L$=6 ($N$ = 18,874,368 cells); the ablations default to $L$=5 ($N$ = 2,359,296 cells) unless noted. All runs use beam width 8, the ViT-B/16 backbone of Sec. 3.4 with roll augmentation (Sec. 3.5), and the optimizer settings of Sec. 3.5. Training uses a single RTX 6000 Ada (48 GB) in fp32 with gradient checkpointing, shared with other resident jobs; the headline $L$=6 model trains in about 12 hours for 60k steps. The per-image inference timings in Tab. 5 are measured separately on the idle GPU.

### 4.2. SYMSOL I: rotation distributions

Tab. 2 reports average log-likelihood per shape on SYMSOL I against the standard baselines. A single joint model over all five shapes reaches 7.71 nats, exceeding the strongest RGB-only baseline (SpyroPose without keypoints, 7.33, which trains one model per shape) and improving on the joint grid baselines by a wide margin (IPDF 4.10, I2S 3.41). Part of this margin may owe to the backbone—we use an ImageNet-21k ViT-B/16 where the baselines use ResNet-50—which Tab. 13 isolates. The full keypoint-based SpyroPose is stronger still (8.97) but relies on 3D CAD keypoints and the authors' own re-render, so it is outside the RGB-only regime (caption). The two NeST rows are separate models trained at $L$=5 and $L$=6; the finer-resolution model gains about a nat (6.68 → 7.71), and the pure resolution effect at fixed model is isolated in Tab. 6.

The per-shape pattern is informative when read against the discretization ceiling in Tab. 1. The continuous shapes come closest to their ceilings: cyl reaches 7.10 of a possible 7.89 and cone 7.56 of 8.58, both within about a nat. The discrete shapes leave substantially more headroom—tet and cube score 9.14 and 9.05 against ceilings near 12 and 11 (gaps of 2.8 and 2.2 nats)—and the icosahedral object is the clear outlier at 5.69, the lowest of the five and furthest below its ceiling (10.37, a 4.7-nat gap). Its 60 equivalent modes force the density to divide its mass among many well-separated peaks at once, which is the binding difficulty here—not the continuous rotational manifolds, which the model represents to within about a nat of the grid limit.

### 4.3. SYMSOL II: dynamic ambiguity

Tab. 3 reports SYMSOL II, where ambiguity depends on marker visibility. [TODO: Run: the three marked shapes (tetX, cylO, sphereX) have been re-dumped and prepared at $L$=6 (data/symsol_II_*); 45k train and 5k test images each. What remains is three per-shape training runs (prior work trains one model per marked shape). Note on evaluation: in this symmetric_solids tfds release every marked-shape test image carries a single ground-truth rotation (verified: $K$=1 for all 5,000 test images of each shape, versus $K$=12/24/60/360/720 for SYMSOL I), so the SYMSOL II metric reduces to log-likelihood at the single annotated pose and the equivalent-set machinery is inactive here. Analysis once numbers exist.]

### 4.4. Pose accuracy

Although NeST is a distribution model, its beam-search MAP estimate yields a point prediction. Tab. 4 reports geodesic error and thresholded accuracy on SYMSOL I. Because the MAP estimate is a cell centre, its error is lower-bounded by the cell-centre quantization of the grid—about 0.45° at $L$=6—so the pose metric saturates at the grid resolution and should be read against that floor. For the continuous and low-order shapes the MAP estimate sits essentially at that floor: cone and cyl reach median error 0.51° and 0.49° with acc@2° > 0.99, and tet and cube follow at 0.6–0.7° with acc@2° of 0.93–0.96. The icosahedral object is again the outlier (median 1.71°, mean 2.58°, acc@2° = 0.579). Because the error is measured to the nearest equivalent ground truth, this is not mode confusion—landing on any of the 60 symmetric copies would score near 0°—but within-mode localization: with the density mass divided across those copies, each per-mode peak is broad, so the argmax leaf lands a few cells off the true pose. The per-level decomposition (Tab. 12) is consistent with this, icosa paying the largest penalties at the finest levels. The errors stay well below the ≥ 72° spacing between icosahedral modes, hence the recovery by acc@10° (0.975); the same broad-peak cause underlies icosa's low log-likelihood in Tab. 2. The overall median of 0.61° confirms that, icosa aside, the MAP estimate is accurate

Table 2. SYMSOL I rotation distribution estimation—average log-likelihood ↑ (nats), per shape and overall, on the original SYMSOL I dataset. Uniform scores −2.29. "Model" is joint (one model over all five shapes) or per-shape; "$L$" is the evaluation grid level ($N{=}72\cdot 8^L$), or "–" for methods without a rotation grid. Baseline numbers are as reported by [17] and [6]. SpyroPose's full keypoint-based method is stronger (7.67/6.96/11.04/10.10/9.06, avg. 8.97) but uses 3D CAD keypoints and the authors' own SYMSOL I re-render, so it is not directly comparable to RGB-only methods on the original dataset.

| Method | Backbone | Model | $L$ | tet. | cube | ico. | cone | cyl. | avg. |
|---|---|---|---|---|---|---|---|---|---|
| Prokudin et al. [19] | – | per-shape | – | −1.86 | −0.50 | −2.39 | −3.34 | −1.28 | −1.87 |
| Gilitschenski et al. [4] | – | per-shape | – | −2.29 | −2.29 | −2.29 | 3.84 | 0.88 | −0.43 |
| Deng et al. [2] | – | per-shape | – | 0.27 | −4.44 | −2.45 | 0.16 | −0.95 | −1.48 |
| IPDF [17] | ResNet-50 | joint | 5 | 5.70 | 4.81 | 1.28 | 4.45 | 4.26 | 4.10 |
| I2S [10] | ResNet-50 | joint | – | 4.78 | 3.27 | 2.15 | 3.75 | 3.10 | 3.41 |
| HyperPosePDF [8] | – | joint | – | 7.04 | 6.77 | 5.10 | 5.74 | 4.73 | 5.78 |
| SpyroPose w/o KP, IS [6] | ResNet-50 | per-shape | 6 | 6.04 | 6.23 | 3.12 | 6.77 | 6.07 | 5.65 |
| SpyroPose w/o KP [6] | ResNet-50 | per-shape | 6 | 8.69 | 8.63 | 5.23 | 7.62 | 6.46 | 7.33 |
| NeST (ours) | ViT-B/16 | joint | 5 | 8.00 | 7.73 | 5.12 | 6.55 | 6.00 | 6.68 |
| NeST (ours) | ViT-B/16 | joint | 6 | **9.14** | **9.05** | **5.69** | 7.56 | **7.10** | **7.71** |

Table 3. SYMSOL II—average log-likelihood ↑ (nats). A separate model is trained per marked shape, following prior work.

| Method | sphX | cylO | tetX | avg. |
|---|---|---|---|---|
| IPDF [17] | 7.30 | 6.91 | 8.49 | 7.57 |
| I2S [10] | 3.74 | 5.18 | 5.61 | 4.84 |
| HyperPosePDF [8] | 7.73 | 7.12 | 8.53 | 7.72 |
| SpyroPose w/o KP [6] | 9.07 | 9.32 | 9.41 | 9.27 |
| NeST (ours) | – | – | – | – |

Table 4. Pose accuracy on SYMSOL I. Geodesic error (degrees, ↓) of the beam-MAP estimate to the nearest equivalent ground truth, and accuracy (↑) at angular thresholds. Overall row averaged over the five shapes.

| Shape | Med.° | Mean° | acc@2° | acc@5° | acc@10° |
|---|---|---|---|---|---|
| tet. | 0.69 | 0.98 | 0.928 | 0.988 | 0.995 |
| cube | 0.60 | 0.92 | 0.960 | 0.981 | 0.987 |
| ico. | 1.71 | 2.58 | 0.579 | 0.900 | 0.975 |
| cone | 0.51 | 0.53 | 0.998 | 1.000 | 1.000 |
| cyl. | 0.49 | 0.50 | 0.999 | 1.000 | 1.000 |
| all | 0.61 | 1.11 | 0.893 | 0.974 | 0.992 |

to the grid floor.

### 4.5. Efficiency

A central claim is that a normalized density costs $O(L)$ rather than $O(N)$. Tab. 5 bears this out. To read the exact, normalized likelihood of a rotation at $L{=}6$, NeST scores a single root-to-leaf path—$L{+}1{=}7$ node evaluations, 5.3 ms/image—whereas IPDF must evaluate its entire 2.36M-cell grid to obtain the same normalization. NeST does this on a grid that is itself 8× finer (18.9M cells), so the saving is more than five orders of magnitude in evaluations at a strictly higher resolution. The beam-MAP decoder adds only the $1{+}bL{=}49$ node scorings needed to return a point estimate (24.0 ms/image). The model is a 114.9M-parameter ViT-B/16 with a peak footprint of 1857 MB, so both modes run comfortably on a single GPU.

Table 5. Efficiency. Node/function evaluations to obtain a normalized density, peak GPU memory, and per-image time (batch size 1). "grid size" is each method's operating resolution ($N{=}72\cdot 8^L$): IPDF must score its full equal-area grid, whereas NeST is evaluated at $L{=}6$ ($N{\approx}18.9$M, a grid 8× finer than IPDF's $L{=}5$ grid) yet scores only one path per rotation for the likelihood and $1{+}bL$ nodes for beam-MAP. Timing and peak memory are measured on an otherwise idle RTX 6000 Ada at batch size 1; peak memory (1857 MB) is measured once over both passes and is dominated by beam-MAP. Baseline memory/time were not measured under our setup ("–").

| Method | grid size | #eval | mem. (MB) | time (ms) |
|---|---|---|---|---|
| IPDF [17] | 2.36M ($L5$) | 2.36M | – | – |
| SpyroPose [6] | pyramid | $\mathcal{O}(k\,8\,L)$ | – | – |
| NeST (likelihood) | 18.9M ($L6$) | $L{+}1{=}7$ | 1857 | 5.3 |
| NeST (MAP, $b{=}8$) | 18.9M ($L6$) | $1{+}bL{=}49$ | 1857 | 24.0 |

### 4.6. Ablations

We ablate the main design choices on SYMSOL I (average log-likelihood unless noted). Further ablations—image backbone, child-scorer conditioning, and the training objective (exact tree-MLE vs. an importance-sampled energy)—are set up in Appendix E.1.

Table 6. Grid depth (resolution transfer). A single level-6 model; its tree-structured density is marginalized (paths truncated) to level $L \in \{3, 4, 5, 6\}$ and scored, with no retraining. Average equivariant-set log-likelihood per shape over 400 test images/shape. The $L$=6 readout (7.72) matches the full test-set headline in Tab. 2 (7.71).

| $L$ | tet. | cube | ico. | cone | cyl. | avg. |
|---|---|---|---|---|---|---|
| 3 | 5.16 | 4.62 | 3.18 | 4.25 | 3.58 | 4.16 |
| 4 | 6.74 | 6.33 | 4.36 | 5.53 | 4.87 | 5.57 |
| 5 | 8.08 | 7.87 | 5.22 | 6.69 | 6.03 | 6.78 |
| 6 | 9.10 | 9.07 | 5.73 | 7.61 | 7.10 | 7.72 |

Grid depth. Our data is dumped at level 6, so a single level-6 model can be read out at any level $L \leq 6$ by truncating paths; Tab. 6 is therefore a resolution-transfer sweep of one model, not a retraining study. Average log-likelihood rises monotonically with depth ($4.16 \to 5.57 \to 6.78 \to 7.72$)—as expected whenever each level's child scorer improves on the uniform split, which holds throughout here, since a finer level then only adds positive log-probability along the path. The per-level gains shrink (+1.41, +1.21, +0.94) while the discretization ceiling of Tab. 1 rises faster ($\approx +1.8$/level), so the gap to the ceiling widens with depth—from 0.43 nats at $L$=3 to 2.30 at $L$=6—i.e. the finer grid offers more headroom than single-rotation training fills. The effect is shape-dependent: the icosahedral object gains least from $L$=5 → 6 (+0.51 vs. +0.9–1.2 for the others), again indicating that its difficulty is mode coverage rather than grid resolution. The $L$=6 readout (7.72 on this subset) matches the full test-set 7.71 of Tab. 2, confirming the subset is representative. A direct check shows the transfer is essentially lossless: on a matched subset, reading the $L$=6 model at $L$=5 (6.78) slightly exceeds a model trained natively at $L$=5 (6.69, itself matching the Tab. 2 row), so training once at the finest resolution and reading out any coarser level costs nothing.

Shared vs. per-level scorer, and beam width. Tab. 7 compares the shared child scorer against one scorer per level, and sweeps the decoding beam width (pose accuracy; headline $L$=6 model, 300 test images/shape). The beam sweep shows clear diminishing returns past $b$=8: tight accuracy improves monotonically with the beam (acc@2° rising 0.845→0.897 from $b$=1 to 16), but the acc@10° tail peaks at $b$=8 (0.991) and regresses at $b$=16 (0.979 overall, and 0.923 on icosa alone). A wider beam retains more distant candidates, which occasionally lets a spurious high-scoring cell far from the true pose win the MAP and inflate the large-error tail on the high-symmetry shape; we therefore use $b$=8 throughout. [TODO: scorer sharing (left column) pending the per-level-scorer training run.]

Table 7. Ablation: scorer sharing and beam width. Left: distribution metric for shared vs. per-level child scorers. Right: pose accuracy (acc@10°) vs. beam width $b$.

| Scorer | avg. LL | $b$ | acc@10° |
|---|---|---|---|
| shared (ours) | – | 1 | 0.982 |
| per-level | – | 4 | 0.990 |
| | | 8 | **0.991** |
| | | 16 | 0.979 |

Table 8. Ablation: training length. Overall SYMSOL I average log-likelihood vs. training steps, from step-tagged checkpoints evaluated on a held-out split carved from the training set (not the test set). Both runs share every hyperparameter; the only difference is roll augmentation. "–" marks steps past where the un-augmented run was stopped after its turnover.

| steps (k) | 10 | 20 | 30 | 35 | 40 | 50 | 60 |
|---|---|---|---|---|---|---|---|
| baseline (no aug.) | 2.98 | 4.39 | **4.84** | 4.60 | – | – | – |
| + roll augmentation | 3.11 | 4.73 | 5.52 | 5.92 | 6.16 | 6.59 | **6.69** |

Training length. Because single-rotation training can overfit which equivalent copy a view was paired with, we track the distribution metric against training length (Tab. 8). Without augmentation the held-out log-likelihood climbs, peaks at step 30k (4.84), and then degrades—by step 35k it has already fallen to 4.60 and is dropping faster each evaluation, the signature of the model memorizing which symmetric copy each view was rendered with rather than learning the shape. Exact in-plane roll augmentation (Sec. 3.5) removes this turnover entirely: the curve rises monotonically to 6.69 at step 60k and is still increasing, with no sign of the memorization collapse. The gain is largest on the high-order shapes—icosahedral held-out LL goes from 2.21 at the baseline peak to 5.08—confirming that the deficit on symmetric shapes was a generalization failure, not a capacity or resolution limit.

### 4.7. SE(3) pose distributions: T-LESS and YCB-Video

We evaluate the SE(3) instantiation of Sec. 3.7 on two real-image BOP datasets [18]. On T-LESS we report the full-SE(3) distribution log-likelihood of the ground-truth pose (Tab. 9), directly comparable to SpyroPose, the only prior method with quantitative SE(3) distribution results; following it we train on physically-based renders (PBR) and evaluate on the real test images

Table 9. SE(3) distribution estimation on T-LESS—average log-likelihood ↑ of the ground-truth pose on the real test images of four representative objects, PBR-trained. Baseline numbers are SpyroPose's real-image results [6]; uniform over the SE(3) grid scores ≈ 2.7. We also report in-box translation coverage.

| Method | obj 1 | obj 14 | obj 25 | obj 27 | avg. |
|---|---|---|---|---|---|
| Uniform [6] | 3.5 | 3.3 | 2.4 | 1.7 | 2.7 |
| SpyroPose w/o IS [6] | 14.1 | 15.9 | 19.2 | 17.2 | 16.6 |
| SpyroPose [6] | 16.9 | 17.5 | 20.8 | 20.0 | 18.8 |
| NeST (ours) | – | – | – | – | – |

Table 10. YCB-Video. Point accuracy of the beam-MAP pose under the BOP metrics (no prior SE(3) distribution baseline exists), with the SE(3) distribution log-likelihood and in-box coverage reported as a reference.

| Method | ADD(-S) ↑ | BOP AR ↑ | avg. LL ↑ |
|---|---|---|---|
| NeST (ours) | – | – | – |

of four representative objects, and size the translation box from the detection (Sec. 3.7). On YCB-Video no prior SE(3) distribution baseline exists, so we report point accuracy under the BOP metrics from the beam-MAP estimate and provide the first SE(3) distribution log-likelihood as a reference (Tab. 10); HB [18] is the natural second distribution benchmark (SpyroPose reports it) and we treat it as an extension. [TODO: Fill both tables: needs the SE(3) head trained on the T-LESS/YCB-V PBR splits (data not yet downloaded; ∼30/100 GB) and the BOP evaluation. Report in-box coverage alongside each likelihood.]

## 5. Conclusion

We presented NeST, a general framework for pose distributions that are exactly normalized by construction: a tree-structured softmax over any strict hierarchical equivolumetric partition of a pose group yields a non-parametric distribution whose log-likelihood at any pose is read off in closed form from a single root-to-leaf path, with no retrieval grid and no sampled partition function. We instantiated it on SO(3) via the Hopf–HEALPix tree and on SE(3) via the Cartesian product of that tree with a bounded equivolumetric translation octree; a single light cross-attention head on a vision transformer realizes both, trained by single-ground-truth maximum likelihood with an $O(bL)$ beam-search MAP decoder.

On SYMSOL I a single joint model attains 7.71 average log-likelihood, above the strongest RGB-only baseline (7.33, trained per shape), while evaluating any rotation's exact likelihood in seven node scorings instead of scoring a million-cell grid. The construction is strongest exactly where normalization is usually most expensive—it comes within about a nat of the discretization ceiling on the continuous shapes (cone, cylinder) and resolves the low-order discrete symmetries well—and it is weakest on the high-order icosahedral object, whose sixty equivalent modes must receive mass simultaneously from single-rotation supervision. The efficiency and the remaining gap share one root cause: an exactly normalized tree makes every path cheap but asks the model to distribute probability consistently across many independent paths, which is the hard part of high-order ambiguity.

Limitations and future work. Three limitations bound the present results. (i) Covering many well-separated modes requires splitting probability mass consistently across independent root-to-leaf paths; this is where single-ground-truth maximum likelihood hurts most, and it is the direct cause of the weaker icosahedral numbers. (ii) The piecewise-constant density caps the attainable log-likelihood at the discretization ceiling of Tab. 1, so headroom on the discrete shapes is a resolution limit as much as a modeling one. (iii) The SE(3) instantiation of Sec. 3.7 is fully constructed and numerically verified—its uniform predictor scores the exact $-\ln(\pi^2 V_{\text{box}})$ floor and its tree likelihood matches the flat per-cell categorical to machine precision—but its evaluation on the real-image benchmarks T-LESS and YCB-Video (Sec. 4.7), together with the SYMSOL II marked-shape evaluation, is the current frontier rather than a reported result. Carrying the verified SE(3) construction through training and evaluation on T-LESS and YCB-Video under the BOP protocol is the main direction we are pursuing.

Reproducibility. The construction admits an exact self-test: a uniform (zero-initialized) predictor scores average log-likelihood $-\ln \pi^2 \approx -2.29$ on SO(3) at every level—and, for the SE(3) instance, $-\ln(\pi^2 V_{\text{box}})$—which we verify numerically before training. The grid is the nested Hopf–HEALPix partition with $N(L) = 72 \cdot 8^L$ cells (Sec. 3.2), fully determined by $L$; evaluation uses the published SYMSOL tfds release. Code and the preprocessing pipeline will be released on acceptance.

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# Exactly Normalized Pose Distributions by Construction: Hierarchical Partition Trees over SO(3) and SE(3)

## Supplementary Material

This supplementary provides the grid construction and its exact self-test (Appendix A), full architecture and training details for reproducibility (Appendices B and C), the decoding procedure (Appendix D), additional results and visualizations (Appendix E), and the planned SE(3) extension (Appendix F).

## A. Grid construction and self-test

Hopf–HEALPix cells. A cell at level $l$ is indexed by a flat integer $c = \text{pix} \cdot n_r(l) + \text{roll}$, where $\text{pix} \in \{0, \dots, 12\,\text{nside}^2 - 1\}$ is a HEALPix pixel in the nested scheme and $\text{roll} \in \{0, \dots, n_r(l)-1\}$ with $n_r(l) = 6\,\text{nside}$ and $\text{nside} = 2^l$. The cell-centre rotation is the Hopf lift of the base point $(\theta, \phi) = \text{pix2ang}(\text{nside}, \text{pix})$ and fibre angle $\psi = (\text{roll} + \frac{1}{2}) \cdot 2\pi / n_r(l)$, giving the ZYZ-Euler rotation $R = R_z(\phi) R_y(\theta) R_z(\psi)$. The inverse (rotation $\to$ cell) reads the base point from $R\hat{z}$ and recovers $\psi$ after removing it; both maps are exact inverses on cell centres.

Strict child structure. In the nested scheme a pixel's four children are $\text{pix}\cdot 4 + \{0, 1, 2, 3\}$ and a fibre bin's two children are $\text{roll}\cdot 2 + \{0, 1\}$, so a cell's eight children are indexed by $\text{child} = \text{s2child}\cdot 2 + \text{rollchild}$. Hence the path $(c_0, c_1, \dots, c_L)$ with $c_0 \in \{0, \dots, 71\}$ and $c_l \in \{0, \dots, 7\}$ uniquely identifies a finest cell, and the parent and the eight children of any node are computed in closed form.

Self-test. The whole likelihood path is verified independent of any training: (i) the cell$\to$centre$\to$cell round trip is exact at every level; (ii) decomposing a flat cell into $(c_0, \text{children})$ and recomposing returns the same cell; and (iii) a uniform predictor ($\pi_0$ and all $\pi_n$ uniform) yields average log-likelihood exactly $-\ln \pi^2 \approx -2.2899$ nats at every level 0–$L$, matching Eq. (5). [TODO: optionally report the measured maximum deviation of the uniform self-test.]

## B. Architecture details

Backbone. ViT-B/16 (vit_base_patch16_224.augreg2_in21k_ft_in1k) [3, 22], $224 \times 224$ input, ImageNet mean/std normalization. We take the forward_features patch tokens and drop the prefix (CLS/register) tokens, giving $P$=196 tokens of dimension $D$=768.

Cross-attention blocks. Both the level-0 scorer and the shared child scorer use the same block: queries are layer-normed and cross-attend (8 heads) to layer-normed patch tokens (key/value), followed by a residual MLP with hidden width $4D$ and GELU; two such layers; a final LayerNorm and linear-to-1 projection. The projection weight and bias are zero-initialized, so each scorer outputs the uniform log-categorical at initialization.

Queries. The level-0 scorer uses 72 learned query embeddings. The child scorer builds its 8 queries from, for each child $j$: the child's offset from the parent centre as a rotation vector $\in \mathbb{R}^3$ normalized by $(\pi^2/N(l))^{1/3}$, and the flattened parent rotation $\in \mathbb{R}^9$; each is Fourier-encoded with $n_{\text{freq}}$=3 (frequencies $\pi \cdot 2^k$, sines and cosines, plus the raw value), concatenated, linearly projected to $\mathbb{R}^D$ through a 2-layer GELU MLP, and summed with a learned per-level embedding.

Parameter count. The ViT-B/16 backbone accounts for 85.8M parameters and the cross-attention tree head for 29.1M, for 114.9M in total. The head count is independent of the grid depth $L$, since a single scorer is shared across all nodes and levels.

## C. Training details

Table 11. Training hyperparameters.

| | |
|---|---|
| optimizer | AdamW [13], weight decay 0.05 |
| learning rate (head / backbone) | $10^{-4}$ / $10^{-5}$ |
| schedule | one-cycle [20], 5% warmup |
| batch size | 64 |
| training steps | 60k |
| gradient clip | 1.0 |
| precision | fp32 (gradient checkpointing) |
| augmentation | in-plane roll (Sec. 3.5) |
| level $L$ (main/abl.) | 6/5 |
| beam width (eval) | 8 |

Training minimizes the single-rotation NLL of Eq. (6); no symmetry annotations and no equivalent sets are used at training time. The data pipeline precomputes, per sample, the ground-truth path and the per-child geometry (offsets and parent rotations) so the network never calls the grid library during training.

Dumping the dataset at level 6 and slicing paths to the desired $L$ exploits the prefix property of the tree, so all depth ablations reuse one dump. All runs use a single RTX 6000 Ada (48 GB) shared with other resident jobs; the headline $L$=6 model completes its 60k steps (plus the final test eval and per-level decompose) in about 12 hours end to end, i.e. $\approx$1.4 steps/s at batch size 64.

## D. Decoding

Beam-MAP proceeds top-down: keep the top-$b$ level-0 cells by $\log \pi_0$; at each level expand every kept cell to its 8 children, score them with the shared child scorer, form cumulative log-probabilities, and keep the top-$b$ cells overall; after level $L$, return the cell-centre rotation of the highest-scoring leaf. Total node scorings $O(bL)$. [TODO: report sensitivity of pose accuracy to $b$ (cross-ref Tab. 7).]

## E. Additional results

Per-shape learning curves. [TODO: figure: training NLL and held-out average log-likelihood vs. steps, per shape; expected to show the discrete high-symmetry shapes lagging.]

Qualitative distributions. [TODO: figure: predicted SO(3) distributions (Hopf/Mollweide visualization) for one example per shape, with the ground-truth symmetry set overlaid; contrast cone/cylinder (continuous) with icosahedron (60 discrete modes).]

Per-level breakdown. Tab. 12 reports the average log-likelihood decomposed by the contribution of each tree level (how much log-probability each level adds along the ground-truth path, on 500 test images/shape from the headline $L$=6 model), to localize where mass is lost. The path log-likelihood is the row sum plus the density constant $\ln(N/\pi^2) = 14.46$. Two patterns stand out. The coarsest level $\ell_0$ carries by far the largest penalty for the high-symmetry shapes ($-4.00$ for icosa, $-3.23$ for cube) because the top-level categorical must divide mass among the many occupied cells of a large symmetry set, whereas the continuous shapes pay least there ($-1.80$/$-2.50$ for cone/cyl). At the finest levels, icosa again pays the most ($-1.21$, $-1.57$ at $\ell_5, \ell_6$ versus $-0.73$, $-1.06$ for tet and $-0.54$, $-0.90$ for cube), i.e. its per-mode peaks stay broad down to the grid floor—the same effect that drives its weaker point accuracy (Sec. 4.4).

SYMSOL II data note. The raw symmetric_solids tfds release (v1.0.0) already contains all eight shapes—the three marked shapes sphereX, cylO and tetX in addition to the five SYMSOL I shapes—with both the single rotation label and the full rotations_equivalent set (train 8 × 45k, test 8 × 5k). Only our preprocessed path-dump was restricted to the five SYMSOL I shapes. The three marked shapes have now been re-dumped and prepared at $L$=6 through the same path-precompute pipeline (data/symsol_II_{tetX,cylO,sphereX}, 45k/5k train/test each); producing Tab. 3 needs the three per-shape training runs (one model per marked shape, the prior-work protocol). Unlike SYMSOL I, each marked-shape test image in this release carries a single ground-truth rotation—we verified $K$=1 for all 5,000 test images of every marked shape (SYMSOL I has $K$=12–720)—so the SYMSOL II metric is log-likelihood at that single pose and no equivalent set enters evaluation. [TODO: Document the exact split and seed used for the per-shape runs.]

Table 12. Per-level contribution to the ground-truth path log-probability (SYMSOL I, headline $L$=6 model, 500 images/shape). $\ell_0$ is the coarsest categorical; $\ell_1 - \ell_6$ are the per-child categoricals. Row sum plus $\ln(N/\pi^2) = 14.46$ equals the path log-likelihood.

| Shape | $\ell_0$ | $\ell_1$ | $\ell_2$ | $\ell_3$ | $\ell_4$ | $\ell_5$ | $\ell_6$ |
|---|---|---|---|---|---|---|---|
| tet. | −2.54 | −0.12 | −0.11 | −0.28 | −0.49 | −0.73 | −1.06 |
| cube | −3.23 | −0.11 | −0.07 | −0.19 | −0.37 | −0.54 | −0.90 |
| ico. | −4.00 | −0.44 | −0.15 | −0.44 | −0.90 | −1.21 | −1.57 |
| cone | −1.80 | −0.71 | −0.72 | −0.74 | −0.80 | −0.93 | −1.20 |
| cyl. | −2.50 | −0.71 | −0.72 | −0.73 | −0.78 | −0.91 | −1.03 |

### E.1. Additional ablations

Beyond the depth, scorer-sharing/beam, and training-length ablations in the main paper (Sec. 4.6), we set up three further ablations on SYMSOL I, reported here as the runs complete.

Backbone. To isolate the distribution formulation from the encoder, we match the backbone to the baselines (Tab. 13); head and protocol are held fixed.

Table 13. Ablation: backbone. Overall SYMSOL I average log-likelihood with different image encoders (head and protocol fixed).

| Backbone | avg. log-lik. |
|---|---|
| ResNet-50 | – |
| ViT-B/16 (in21k→in1k) | – |

Child-scorer conditioning. The child scorer is conditioned on the normalized child offset, the Fourier-encoded parent rotation, and a level embedding. Tab. 14 removes each in turn.

Table 14. Ablation: child-scorer conditioning. Overall SYMSOL I average log-likelihood as conditioning components are removed.

| Conditioning | avg. log-lik. |
|---|---|
| full (offset + parent + level) | – |
| – child offset | – |
| – parent rotation | – |
| – level embedding | – |
| – Fourier encoding (raw inputs) | – |

Training objective. A central claim is that the tree can be trained directly by exact maximum likelihood, rather than as an energy with a sampled partition function. Keeping the head fixed, Tab. 15 compares our tree-MLE objective against an InfoNCE objective with importance-sampled negatives drawn from the tree (the SpyroPose-style training), which SpyroPose reports trains more stably than a directly normalized likelihood.

Table 15. Ablation: training objective. Same head and backbone; exact tree-MLE (ours) vs. an InfoNCE objective with tree-importance-sampled negatives. Overall SYMSOL I average log-likelihood, and negatives scored per step.

| Objective | #neg./step | avg. log-lik. |
|---|---|---|
| InfoNCE + importance sampling | – | – |
| exact tree-MLE (ours) | 0 | – |

## F. SE(3) construction and self-tests

The SE(3) grid of Sec. 3.7 is implemented as a product of the rotation tree and a translation octree. The octree lives in a normalized unit cube $g \in [-\frac{1}{2}, \frac{1}{2}]^3$, subdivided into the regular $2^l \times 2^l \times 2^l$ grid; a cell is addressed by its integer coordinates and, equivalently, by an octant path (octant $= b_x + 2b_y + 4b_z$, most-significant bit first). The affine map $t = \hat{t} + Ag$ carries the cube into the camera frame; since it is affine it preserves equal volume, so $V_{\text{box}} = |\det A|$ and each cell has volume $|\det A|/8^l$. A joint cell is the pair of a rotation path and a translation path, and its per-node categorical is the 64-way product (child $= c_{\text{rot}} \cdot 8 + c_{\text{oct}}$) while both halves subdivide, 8-way afterwards, giving independent depths $L_r, L_t$.

Before any data or training we check four identities on the implementation. (i) Octree cell→centre→cell is exact at every level. (ii) A cell's octant path recomposes the same cell. (iii) A uniform predictor yields average log-likelihood exactly $-\ln(\pi^2 V_{\text{box}})$ at every pair $(L_r, L_t)$, matching Eq. (7) (maximum deviation $\sim 10^{-15}$ nats over a range of depths and a non-trivial affine box). (iv) The tree-structured softmax of empirical per-node conditionals equals the flat empirical joint categorical to machine precision, confirming the factorization and the density constant end to end without a network. The box bounds come from a detection, as in [6]; a pose whose translation falls outside the box has infinite NLL, so we report the in-box coverage rate alongside the likelihood. [TODO: Report T-LESS/YCB-V numbers (Tabs. 9 and 10) and the training details for the SE(3) head once the real-image runs complete.]